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MCQ MASTER SHORTCUTS — BY THUNDERSTUDY

Used for random events when the average (λ) is known.

Counting occurrences (0, 1, 2, 3...)



Main Formula & Identification

$$P(X=x) = (e^{-\lambda} \cdot \lambda^x) / x!$$

Shortcut: Find λ first. Words denoting λ : "Average", "Mean", "Rate per unit".



Trick 1: Memorize $e^{-\lambda}$ Values (80% Time Saver)

λ (Average)	$e^{-\lambda}$ (Approx)
1	0.37
2	0.14
3	0.05
4	0.018
5	0.007

Trick 2: $x = 0$ and $x = 1$ Shortcuts

No Event ($x=0$):

$$P(0) = e^{-\lambda}$$

(No factorial, no power!)

Exactly One ($x=1$):

$$P(1) = \lambda \cdot e^{-\lambda}$$

Trick 3: Ratio Method (Biggest Time Saver)

Instead of full formula:

$$P(x+1) / P(x) = \lambda / (x + 1)$$

If $\lambda=2$, $P(1) = 2 \times P(0)$, $P(2) = (2/2) \times P(1)$. No formula needed, just multiply!



Trick 4: "At Least" & "At Most"

- **At least 1:** $P(X \geq 1) = 1 - P(0) = 1 - e^{-\lambda}$
- **At most 2:** $P(0) + P(1) + P(2)$
- **At least 2:** $1 - P(0) - P(1)$



10-Second MCQ Solving Strategy

1. Find λ (Average).
2. Check required x value.
3. Start from $P(0)$ using memorized values.
4. Use the **Ratio Trick** to move from $P(0)$ to $P(x)$.
5. Spot multiples of $e^{-\lambda}$ in the options for an instant answer.



ULTRA MEMORY TRICK

"Zero $\rightarrow$ e power minus lambda,
Next $\rightarrow$ multiply lambda,
Then divide step by step."

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